

Motor Insurance Tariff Adequacy

A Monte Carlo Case Study Based on a
Compound Poisson–Gamma Model

Simulation Methods in Finance and Insurance – HEC Lausanne

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Executive Summary

The motor insurance portfolio currently operates at a **combined ratio of 0.92**, meaning that for every dollar collected in premiums, only 92 cents are spent on claims and expenses. This translates into an average underwriting surplus of approximately **\$96.73 per policy**, or 7.8% of the premium charged, confirming that the portfolio is profitable under the current tariff.

Based on our analysis, a **tariff reduction of up to 7%** appears commercially feasible without compromising profitability. Beyond this threshold, the portfolio would begin to generate underwriting losses. A reduction in the range of 3-5% is recommended as a prudent starting point, preserving an adequate safety margin while supporting the marketing department’s competitiveness objectives. It is worth noting that SUV policies show lower tolerance for tariff reductions, becoming unprofitable after approximately a 3% cut, and should therefore be treated separately in any repricing exercise.

From a risk perspective, the Solvency Capital is **\$72,098**, and while reinsurance is not strictly necessary at this stage, an Excess of Loss structure covering losses beyond **\$1.16 M** would provide additional protection against extreme claim years and is worth considering as a prudent risk management measure.

1 Introduction

Insurance companies continually seek ways to remain competitive while preserving financial sustainability. The product development department plays a central role in this balance, designing and pricing policies that are both attractive to customers and aligned with the company’s profitability targets. In the current market environment, the marketing department has expressed interest in reducing the motor insurance tariff to strengthen the company’s competitive position.

Using a portfolio of 1,002 motor insurance policies, we build statistical models for both the frequency and severity of claims, simulate the aggregate loss distribution via Monte Carlo methods, and derive key risk and profitability metrics. The analysis follows a structured framework: data exploration and cleaning, model selection and validation, risk premium estimation, Value-at-Risk computation under the Solvency II standard, and a sensitivity analysis of the combined ratio under different tariff reduction scenarios.

The findings are intended to provide the marketing department with a clear, evidence-based answer to the question: *by how much, if at all, can the current tariff be safely reduced?*

2 Data Analysis

2.1 Dataset Overview

The raw dataset contains 1002 rows and 15 columns. Each row represents one insurance policy. The columns include the number of claims (**CLM_FREQ**), six claim amount fields (**CLM_AMT_1** through **CLM_AMT_6**), the annual premium (**PREMIUM**), administrative fees (**FEES**), policyholder age (**AGE**), and categorical variables for gender (**GENDER**), vehicle type (**CAR_TYPE**), vehicle use (**CAR_USE**), and geographic segment (**AREA**).

Summary statistics for the key numerical variables are presented in Table 1.

	CLM_FREQ	PREMIUM	FEES	AGE
Min.	0.000	1000	50.00	19.00
1st Qu.	1.000	1117	56.00	39.00
Median	1.000	1240	62.00	44.00
Mean	1.559	1241	62.07	44.57
3rd Qu.	2.000	1361	68.00	51.00
Max.	6.000	1499	75.00	80.00

Table 1: Summary statistics for key portfolio variables.

2.2 Data Cleaning

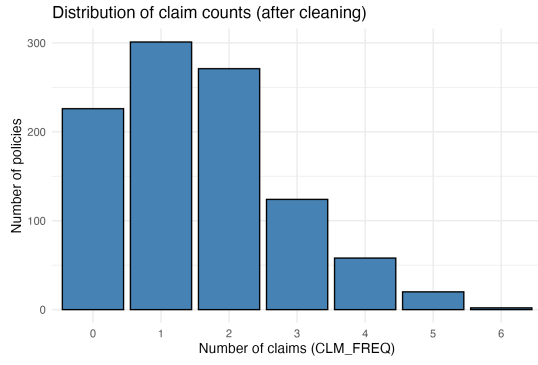
Due to the data being stored with mixed formats a cleaning process was required. The claim amount columns were standardized to a numeric format and two records were excluded since they presented anomalies. Both records had negative claim amounts and one additionally presented an extreme claim value that was 50 times larger than the second-largest claim recorded. No inconsistency was found in the categorical variables.

2.3 Exploratory Analysis

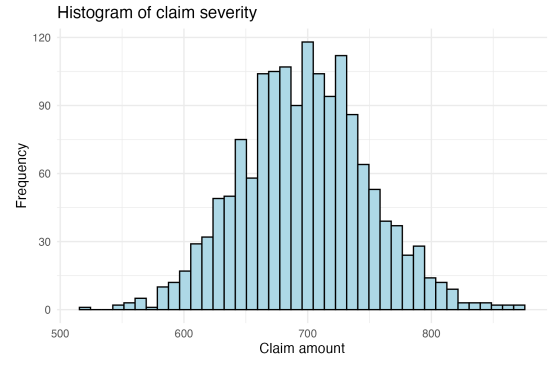
The distribution of claim counts across 1,002 policies is presented in Figure 1a. A large proportion of policyholders (22.6%) filed no claims, while the majority filed one or two claims. Very few policies reached five or six claims, which is typical of motor insurance portfolios.

The mean number of claims per policy is 1.56 and the variance is 1.57, giving a variance-to-mean ratio of 1.012. Since the Poisson Distribution assumes equality between the mean and variance, this result provides a first indicator that a Poisson model may be suitable for modelling claim frequency. Figure 1a confirms that most policies generate between one and two claims, with very few extreme observations.

On the other hand, the histogram of claim severity (Figure 1b) exhibits an approximately bell-shaped and symmetric distribution centred around \$700, with a relatively narrow spread. The near symmetry of the distribution and the limited variability suggest that both the Gamma Distribution and the Normal Distribution may be plausible candidates for modelling claim severity.



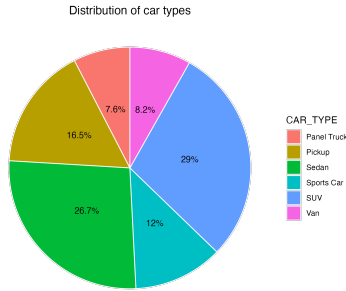
(a) Distribution of claim counts.



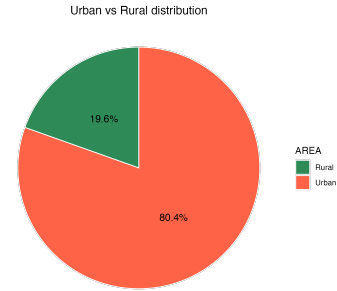
(b) Histogram of claim severity.

Figure 1: Empirical distributions of claim frequency and severity.

Moving forward, the portfolio composition was analysed by geographical area and vehicle use. Overall, the portfolio is predominantly composed of policies located in urban areas (80.4%), while rural areas account for the remaining 19.6%.



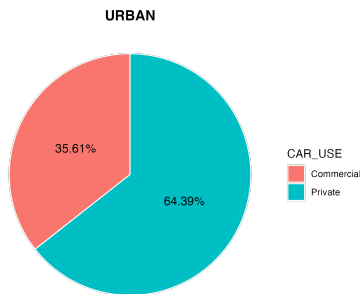
(a) Urban portfolio by car type.



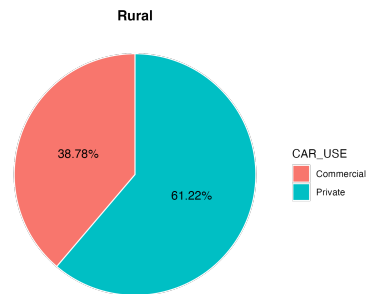
(b) Rural portfolio by car type.

Figure 2: Vehicle type distribution by geographical area.

Regarding vehicle use, the urban portfolio contains a higher proportion of vehicles used for private purposes compared to the rural portfolio, which is consistent with the more agriculture- and field-related activities typically associated with rural areas.



(a) Urban portfolio by car use.



(b) Rural portfolio by car use.

Figure 3: Vehicle use distribution by geographical area.

Across both sectors, SUVs and sedans account for more than half of the insured vehicles, making

them the dominant vehicle categories within the portfolio.

3 Model Selection and Validation

3.1 Frequency of Claims

There are three candidate distributions for claim frequency: the Poisson, Negative Binomial, and Geometric distributions, all fitted by maximum likelihood estimation (MLE). The Akaike Information Criterion (AIC) was used as the primary model comparison metric: a lower AIC indicates a better trade-off between goodness of fit and model complexity.

Distribution	Parameters (MLE)	AIC
Poisson	$\lambda = 1.5559$	3148.21
Negative Binomial	size = 135.67, $\mu = 1.5559$	3150.15
Geometric	$p = 0.3913$	3430.18

Table 2: AIC comparison for claim frequency distributions.

The Poisson model achieves the lowest AIC, slightly better than the Negative Binomial and substantially better than the Geometric.

To formally validate the Poisson fit, a chi-square goodness-of-fit test was conducted. The test statistic is 5.65 on 4 degrees of freedom, yielding a p-value of 0.2269. Since the p-value is well above the 5% significance threshold, the null hypothesis that the data follow a Poisson distribution cannot be rejected. The visual diagnostics also show close alignment between observed and theoretical values, further supporting the Poisson model. Therefore, the chosen frequency model is Poisson ($\lambda = 1.5559$).

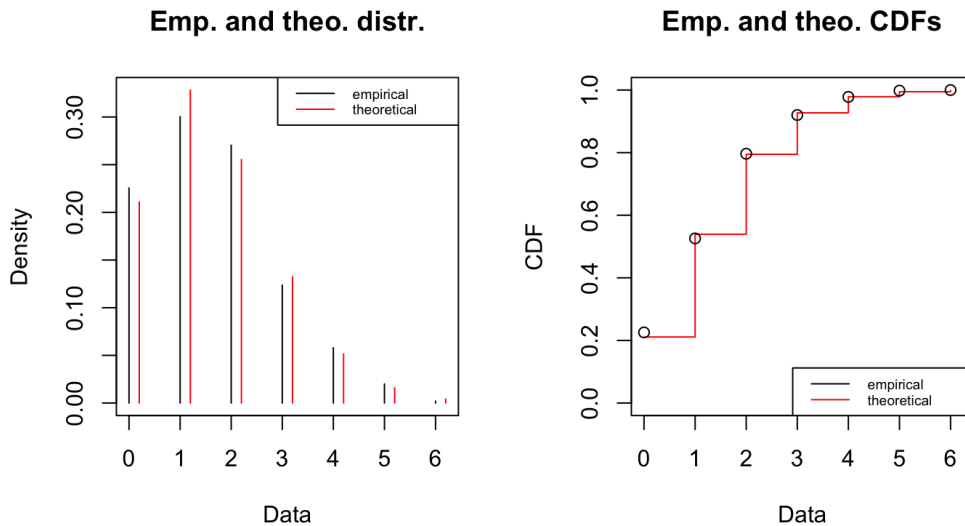


Figure 4: Diagnostic plots for the Poisson fit.

3.2 Severity of Claims

Six candidate distributions were fitted to the 1,559 cleansed individual claim amounts using MLE: Gamma, Lognormal, Weibull, Normal, Exponential, and Pareto.

Distribution	AIC
Gamma	16717.17
Lognormal	16721.05
Weibull	16913.45
Normal	16718.24
Exponential	23535.42
Pareto	23537.42

Table 3: AIC test comparison for claim severity distributions.

Distribution	K-S p-value
Gamma	0.5816
Normal	0.4375
Lognormal	0.6491

Table 4: KS test comparison for claim severity distributions.

The Gamma distribution achieves the lowest AIC and a high K-S test p-value of 0.5816, indicating that the null hypothesis of Gamma distribution cannot be rejected at any conventional significance level. The Normal distribution has a very close performance (AIC = 16,718.24, K-S p-value = 0.4375). The difference in AIC between the two is negligible (1.07), so both provide statistically equivalent fits.

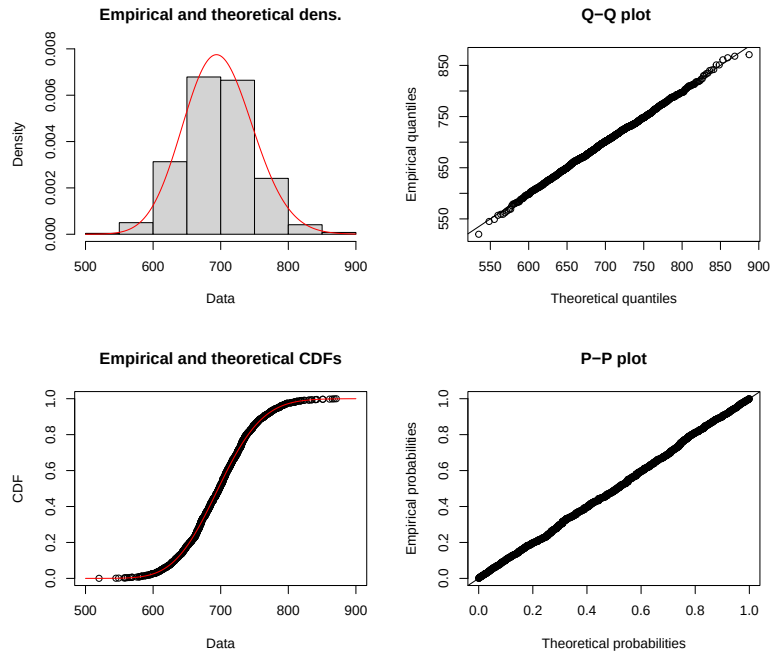


Figure 5: Diagnostic plots for the Gamma fit.

The Gamma distribution is the more appropriate choice since, unlike the Normal, it is defined only on positive values, which corresponds to the domain for claim amounts.

The diagnostic plots confirm excellent fit: the theoretical density closely matches the empirical histogram, and the Q-Q and P-P plots lie almost perfectly on the diagonal, with only minor deviations at the extremes of the distribution.

Therefore, the chosen severity model is Gamma ($\alpha = 182.75$, $\beta = 0.2620$), with an estimated mean of \$697.57 and a standard deviation of \$51.52.

3.3 Monte Carlo Estimation: Frequency

Using the fitted Poisson ($\lambda = 1.5559$) model, $N = 100,000$ draws were simulated and the Monte Carlo (MC) estimator of the expected number of claims was computed. The results demonstrate the Law of Large Numbers in practice:

Metric	Value
MC mean	1.5512
Empirical mean (data)	1.5559
MC variance	1.5524
Theoretical variance	1.5559
Empirical standard error	3.940095×10^{-3}
Theoretical standard error	3.944475×10^{-3}
MSE	1.552435×10^{-5}

Table 5: Monte Carlo estimation results for claim frequency.

The MC mean is extremely close to the empirical mean and the theoretical λ , confirming that the estimator is unbiased. The convergence plot shows that the running mean stabilises rapidly around λ after a few thousand iterations, consistent with the CLT.

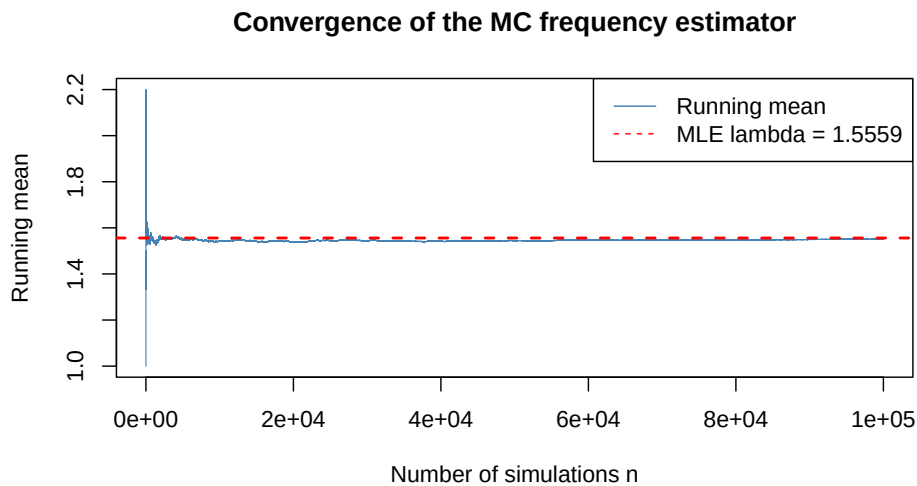


Figure 6: Convergence of the MC frequency estimator.

3.4 Monte Carlo Estimation: Severity

Similarly, $N = 100,000$ draws from the $\text{Gamma}(\alpha, \beta)$ severity model yield the following results:

Metric	Value
MC mean	697.72
Empirical mean (data)	697.57
MC variance	2665.22
Theoretical variance (α/β^2)	2662.65
Empirical standard error	0.16325
Theoretical standard error	0.16318

Table 6: Monte Carlo estimation results for claim severity.

Again, the MC estimator is unbiased: the simulated mean (\$697.72) is within \$0.15 of the empirical mean (\$697.57). The empirical variance (2,665.22) matches the theoretical variance (2,662.65) almost exactly. The convergence plot shows smooth and rapid convergence of the running mean to the true value, with slightly more initial volatility than for frequency due to the continuous nature of the Gamma distribution.

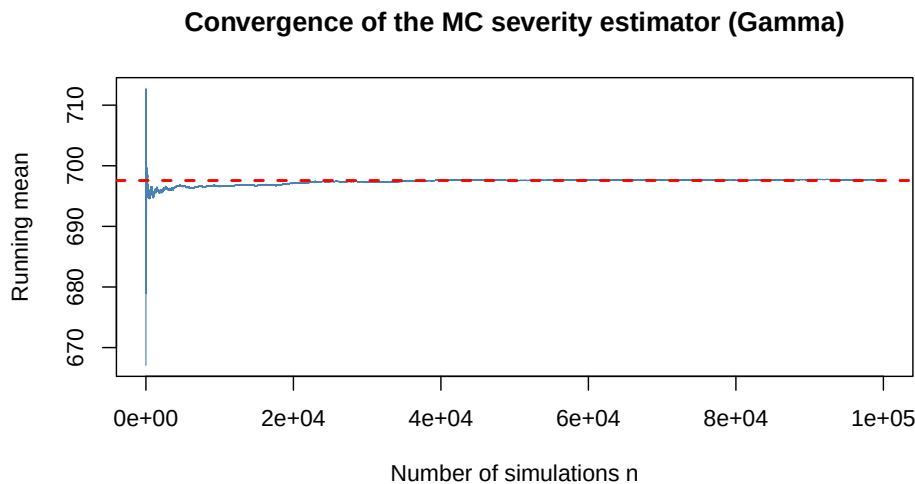


Figure 7: Convergence of the MC severity estimator (Gamma).

3.5 Variance Reduction Techniques

Variance reduction techniques (VRTs) are important in this context because reducing the variance of the Monte Carlo (MC) estimator leads directly to tighter confidence intervals for key quantities such as the risk premium and Value at Risk (VaR). This enables more reliable decisions with fewer simulations. Two variance reduction techniques were implemented:

- **Antithetic variates (AV):** For a set of $n/2$ uniform draws U , each draw is paired with its complement $1 - U$. Because these two are negatively correlated, their average has a much smaller variance than the crude MC estimator.
- **Control Variates (CV):** For severity, the Normal distribution (fitted to the same data)

is used as a control variate. It is almost perfectly correlated with the Gamma severity (large shape parameter), so using it removes most of the variance.

Variable	Technique	Var (Crude)	Var (Reduced)	Var. Reduction	Correlation
Frequency	Antithetic	1.55×10^{-5}	2.10×10^{-6}	86.45%	-0.8641
Severity	Antithetic	2.67×10^{-2}	6.41×10^{-5}	99.76%	-0.9976
Severity	Control Variate	2.67×10^{-2}	3.18×10^{-5}	99.88%	+0.9994

Table 7: Variance reduction results for frequency and severity estimators.

For frequency, the antithetic method reduces variance by 86.45%, driven by a strong negative correlation of -0.8641 between paired draws. For severity, both the antithetic (99.76%) and control variate (99.88%) methods achieve near-total variance elimination. The control variate is marginally more effective because the Gamma and Normal distributions are almost identical at this parameter regime (correlation = 0.9994), making the control almost perfect.

The effectiveness of both methods depends on how closely related the two sets of draws are. For the antithetic method, when one draw is high, its pair tends to be low, and vice versa; this negative correlation (correlation of -0.8641 for frequency and -0.9976 for severity) cancels out much of the randomness. For the control variate method, the Gamma severity draws and the Normal draws generated from the same uniform inputs are almost perfectly correlated (0.9994), so the Normal can be used to correct the Gamma estimate and remove nearly all remaining uncertainty.

4 Application

This section starts with the computation of the risk premium using two approaches: the traditional empirical method based on observed claims experience, and the Monte Carlo simulated mean aggregate loss derived from the collective risk model. This is followed by an analysis of the Value at Risk (VaR) under the Solvency II framework and its implications for the company’s financial sustainability. Finally, the combined ratio is examined as a profitability indicator to assess the adequacy of current underwriting performance. Based on these results, it is evaluated whether reinsurance is necessary to protect capital, stabilize earnings, and reduce catastrophe risk without unduly harming long-term profitability.

4.1 Risk Premium

Actuarial Premium

The following equation illustrates the actuarial risk premium, which represents the expected claims cost per policyholder and corresponds to the pure premium required to cover future losses before allowing for expenses, commissions, and profit margins.

$$\pi = \frac{\sum \text{Claim Amount}}{\text{Number of Policies}} \quad (1)$$

Using the available portfolio data, the empirical risk premium was estimated at **\$1,085.34** per policy. This value reflects the average historical claims managed by the insurer and provides a first benchmark for tariff adequacy.

Aggregate Claim Amount Distribution

The second approach to estimating the risk premium consists of calculating the expected aggregate loss per policy through Monte Carlo simulation under the collective risk model. In this approach, the annual total claim amount per policy is defined as

$$S = \sum_{j=1}^N Y_j, \quad N \sim \text{Poi}(\lambda), \quad Y_j \sim \text{Gamma}(\alpha, \beta), \quad N \perp (Y_j) \quad (2)$$

The mean and variance of S are given by

$$\mathbb{E}[S] = \lambda \mathbb{E}[Y] = \lambda \frac{\alpha}{\beta}, \quad \text{Var}[S] = \lambda \mathbb{E}[Y^2] = \lambda \frac{\alpha(\alpha + 1)}{\beta^2} \quad (3)$$

S is estimated using a two-step algorithm drawing N observations from a Poisson, and after simulating N i.i.d. Gammas, and finally adding the values. Using $N = 100,000$ simulated scenarios, we obtain the MC-estimator **\$1,081.19**. The simulated aggregate loss distribution was found to be positively skewed. In practical terms, this means that most policyholders generate no claims or only moderate claim amounts, while a relatively small proportion of cases generate significantly larger losses. Figure 8 presents the simulated aggregate claim distribution.

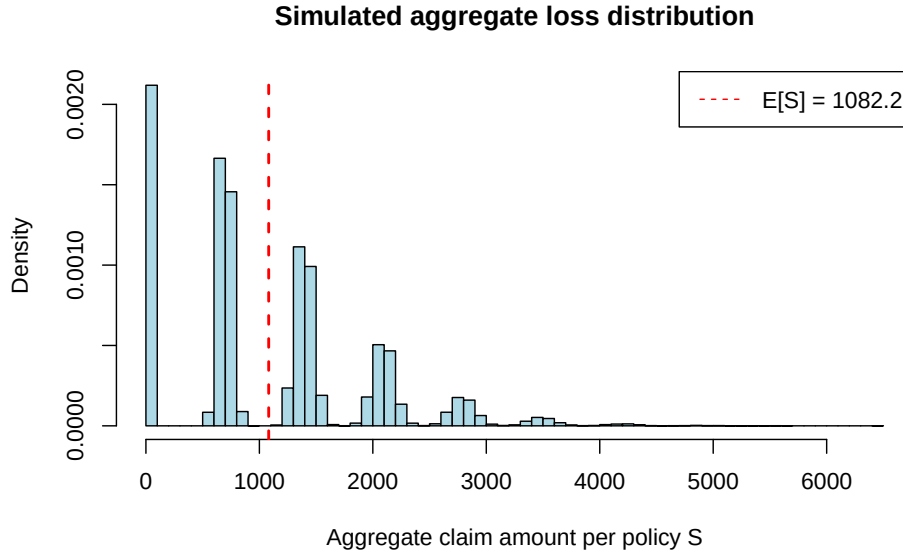


Figure 8: Aggregate loss distribution.

4.2 Comparison of Results

The estimated risk premium is very close to the empirical risk premium previously obtained from observed data, providing reassurance that the selected modelling assumptions are consistent with the historical claims experience of the portfolio. The results can be summarized as follows.

Estimator	Value per policy (\$)
Empirical $\sum_i \text{Total Claims}_i / 1002$	1,085.339
Monte Carlo (Compound Poisson)	1,082.193
Average Premium Charged	1,241.028
Average Fees	62.0720
Margin per policy	96.7623

Table 8: Comparison of risk premium estimators.

The margin can be expressed as

$$\text{Margin} = \text{Premium} - \text{Risk Premium} - \text{Fees}.$$

Using the MC estimator, the expected underwriting surplus is approximately \$96.76 per policy, or about 7.8% of the premium charged. The premium (\$1,241.03) is sufficient to cover expected losses plus fees (\$1,144.26), generating a positive margin. This result suggests that the current tariff structure remains technically adequate and continues to generate a moderate profitability buffer.

4.3 Value-at-Risk under Solvency II Framework

By definition, the VaR at level $\alpha \in (0, 1)$ is the α -quantile for the loss distribution,

$$\text{VaR}_\alpha(S) = \inf\{x : F_S(x) \geq \alpha\} \quad (4)$$

Under the Solvency II regulatory framework, insurers are required to hold sufficient capital so that the probability of insolvency over a one-year horizon does not exceed 0.5%. This corresponds to a confidence level of $\alpha = 0.995$.

The VaR was estimated through the empirical quantile approach, using the generated aggregate loss distribution. This method is appropriate because the simulated loss distribution is positively skewed, making normal approximations less reliable in the tail. The resulting estimates are summarized below.

Method	VaR (\$)
Empirical Quantile (99.5% per policy)	4,002.67
Empirical Quantile (99.5% total portfolio)	1,159,529
Total Expected Losses	1,087,431
Solvency Capital Requirement	72,098

Table 9: VaR and Solvency Capital Requirement estimates.

At portfolio level, the 99.5% VaR was estimated at \$1,159,529, compared with expected total annual losses of \$1,087,431. The difference between these two quantities defines the Solvency Capital Requirement (SCR), i.e. $\text{SCR} = \text{VaR}_{99.5\%} - \mathbb{E}[S]$.

Therefore, the company would need to hold approximately \$72,098 of capital above expected losses in order to comply with the Solvency II solvency requirements and maintain the company's sustainability.

4.4 Reinsurance

Reinsurance is generally used to protect the insurer against unusually large losses, reduce volatility of annual results, and support capital adequacy. The need for reinsurance was evaluated using the combined ratio as a profitability indicator together with the tail risk measures obtained from the aggregate loss simulation. The combined ratio is defined as:

$$CR = \frac{\text{Total Claims} + \text{Administrative Costs}}{\text{Total Premium Paid}} \quad (5)$$

Based on the portfolio data, the estimated combined ratio was 0.9246, which is below 1. This indicates that premium income is currently sufficient to cover claims and expenses, and that the portfolio remains profitable on an underwriting basis.

While reinsurance is not strictly necessary to restore profitability or solvency under the current assumptions, it may still be beneficial for reducing volatility and protecting against extreme losses. A reasonable threshold would be near the 99.5% percentile of total annual losses, corresponding to \$1,159,529. Overall, the company appears capable of retaining its current level of risk, subject to the stability of the underlying assumptions.

5 Model Explanation

5.1 Combined Ratio

The combined ratio measures whether premiums collected are sufficient to cover claims and operating expenses. A Combined Ratio < 1 means the portfolio is profitable while a Combined Ratio > 1 means underwriting loss. The insurance portfolio's current combined ratio is 0.9246. This is a positive result and indicates that the current tariff remains commercially viable while still generating profit.

Impact of Tariff Reductions

A sensitivity analysis was performed to assess how premium reductions would affect profitability.

Tariff Reduction	0%	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
Combined Ratio	0.92	0.93	0.94	0.95	0.96	0.97	0.98	0.99	1.01	1.02	1.03

Table 10: Sensitivity of the combined ratio to tariff reductions.

This suggests that a moderate reduction in rates could be feasible if the objective is to improve competitiveness or increase market share. However, reductions exceeding 7% result in losses on policy underwriting. A safe operating zone lies below 5%, which keeps the combined ratio at or below 0.96 and preserves a reasonable underwriting margin to absorb the cost of regulatory capital (SCR) and unforeseen losses.

Nevertheless, applying a uniform tariff reduction across the entire portfolio may introduce additional risks, such as attracting disproportionately riskier policyholders or reducing profitability in already marginal segments. An alternative and potentially more effective approach is to implement differentiated pricing by vehicle type.

The segmentation analysis revealed that SUVs exhibit the lowest tolerance for tariff reductions, becoming unprofitable after approximately a 3% reduction, as shown in Table 11. In contrast, the remaining vehicle categories, when grouped together, remain profitable even after reductions of up to 9%. These findings suggest that non-SUV vehicles could support more competitive pricing strategies, while SUV tariffs should be adjusted more conservatively in order to preserve an adequate safety margin.

Tariff Reduction	0%	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
SUV	0.96	0.97	0.98	0.99	1.00	1.01	1.02	1.03	1.04	1.06	1.07
Non-SUV	0.91	0.92	0.93	0.94	0.95	0.96	0.97	0.98	0.99	1.00	1.01

Table 11: Sensitivity of the combined ratio to tariff reductions, by vehicle category (SUV vs Non-SUV).

5.2 General Remarks and Conclusions

The key findings are as follows:

- **Pricing adequacy and profitability:** The portfolio is profitable on average. The current combined ratio is 0.92, indicating that premiums cover claims and expenses. In addition, the expected underwriting margin is approximately \$96.76 per policy (around 7.8% of premium income).
- **Tariff flexibility:** A uniform tariff reduction of up to 7% is technically feasible, however a safety margin should be considered. A higher reduction would lead to underwriting losses.
- **Reinsurance position:** Reinsurance is optional rather than urgent under the current portfolio, as it is profitable and can absorb expected risks. However reinsurance could still be useful: an Excess of Loss contract would protect the company against extreme scenarios while allowing normal losses to remain fully retained. Such a structure would preserve most underwriting profitability while reducing exposure to catastrophic or highly adverse claim years.

Overall, the recommended strategy would be to consider a measured tariff reduction in order to improve competitiveness, while maintaining regular monitoring of claims trends, profitability, and capital adequacy.